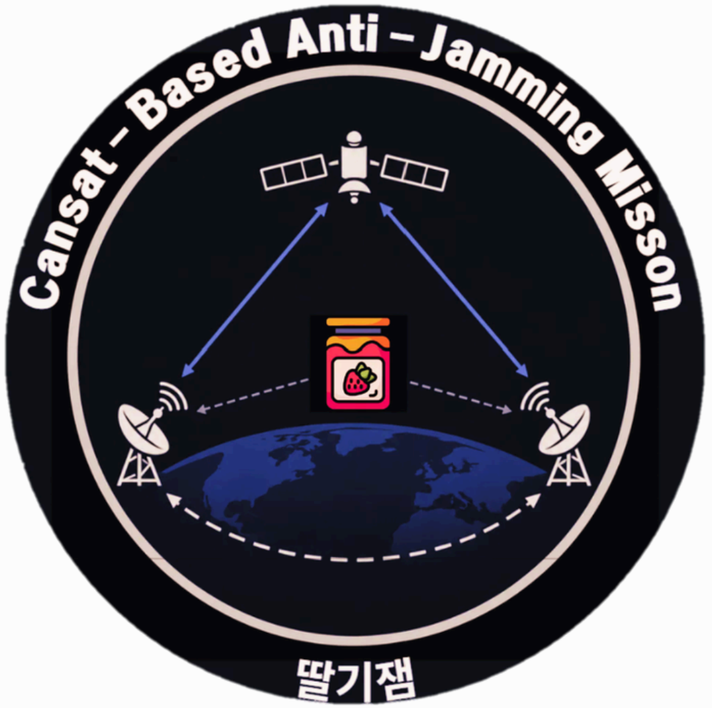
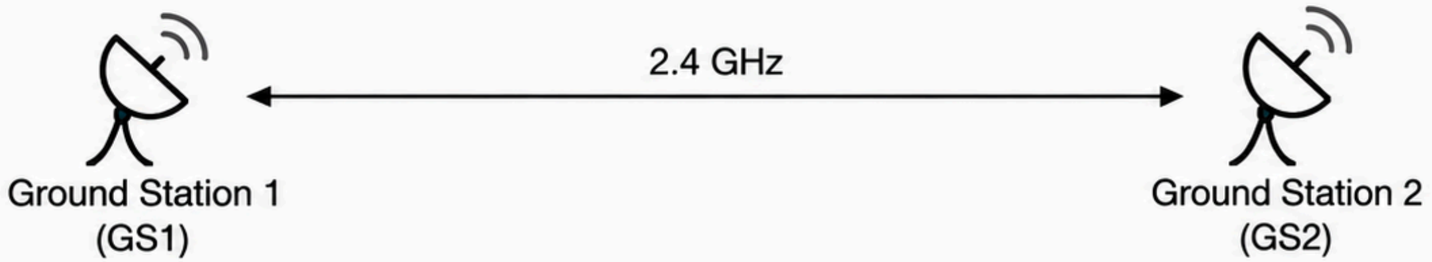
DDALGI - JAM



A Study on Anti-Jamming and Telecommunication Recovery Based on Frequency Diversity Using CanSat Relays

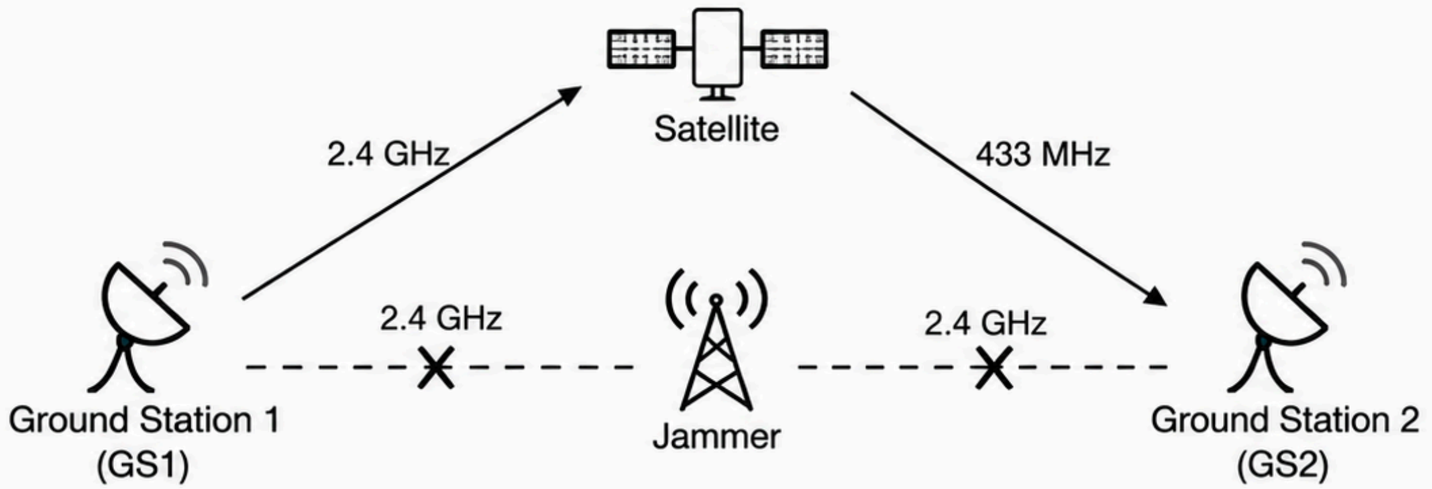
Mission

1. Normal Communication



- GS1 and GS2 communicate directly over 2.4 GHz.
- The satellite is not used.
- No jamming is present.

2. Anti-Jamming Communication

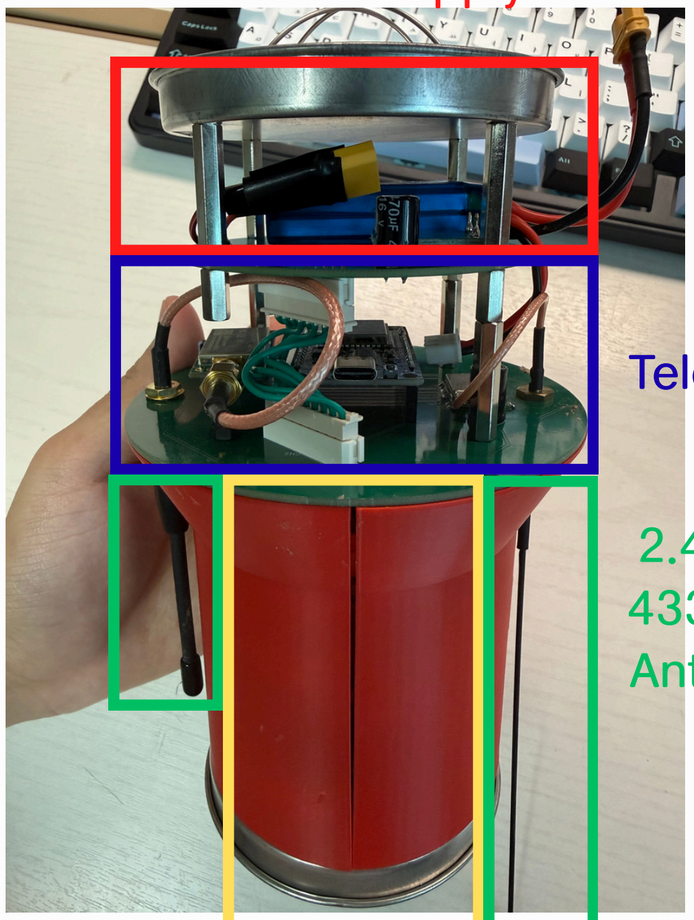


- GS2 is jammed on 2.4 GHz, so direct communication fails.
- GS1 transmits to the satellite on 2.4 GHz.
- The satellite forwards the signal to GS2 on 433 MHz.

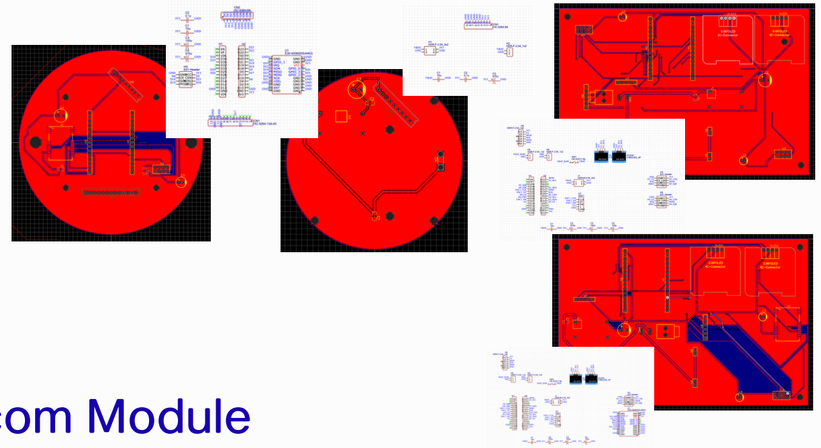
DDalgi_Jam

Hardware Structure

PowerSupply



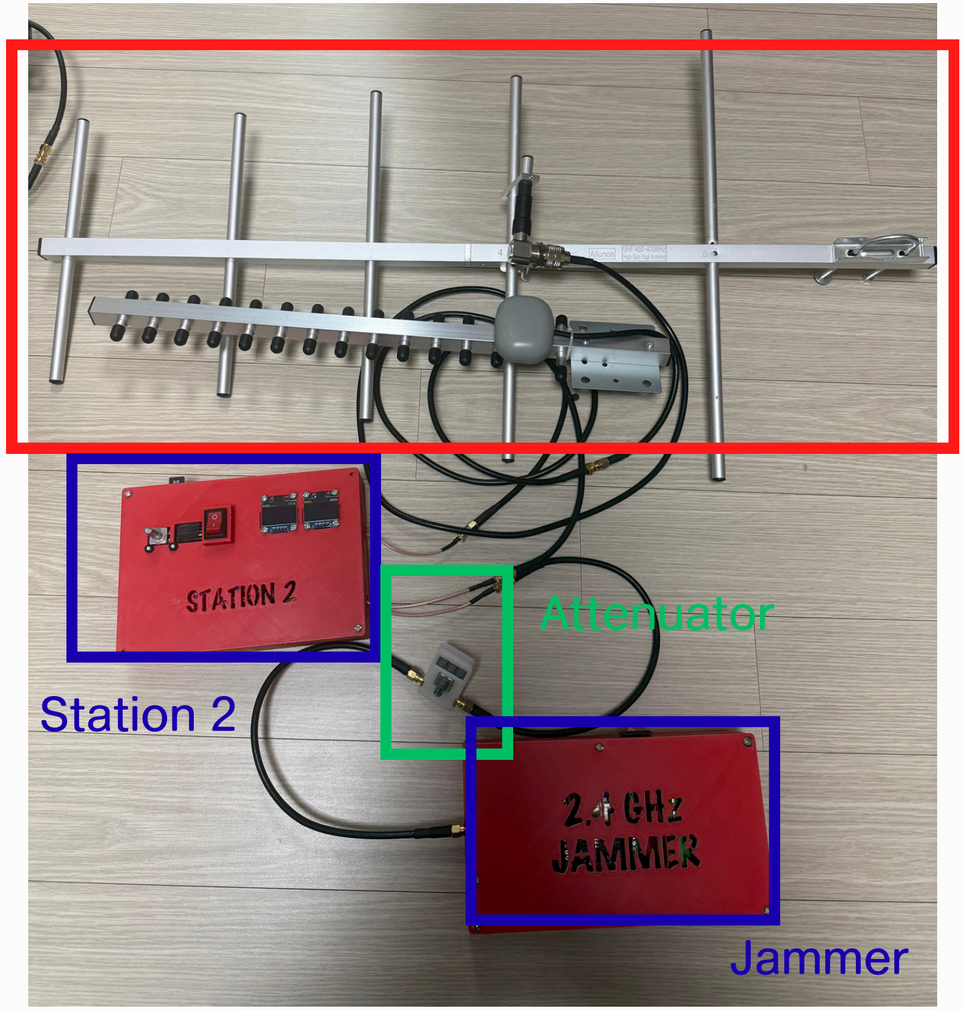
Telecom Module



2.4 Ghz
433 Mhz
Antenna

Existing Cansat Kit

Station 2.4, 433 Yagi

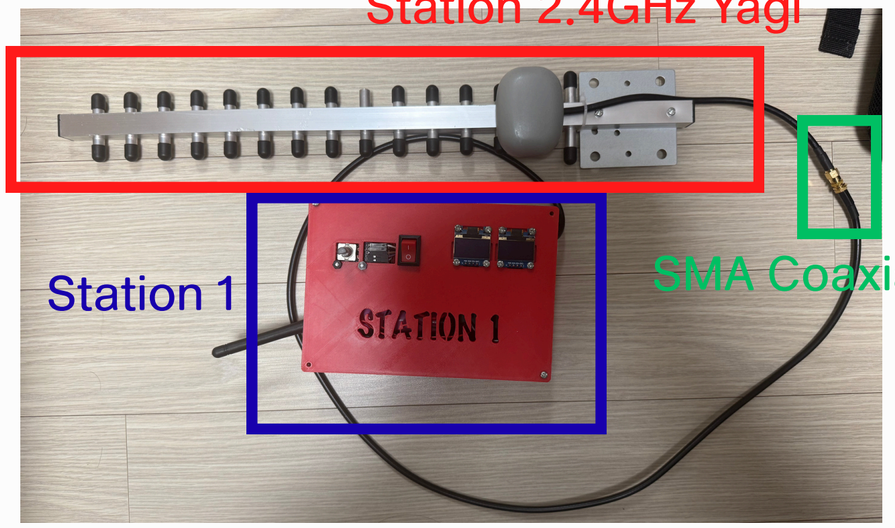


Station 2

Attenuator

Jammer

Station 2.4GHz Yagi



Station 1

SMA Coaxial Cable

Software Structure



$$J_{ch} = 0.6 \times (1 - PDR) + 0.4 \times RPD_{duty}$$

01 (1−PDR): packet loss rate, computed from seq values

02 RPD_duty: RF energy detection duty cycle

Jch 〉 0.6 sustained for 0.5 s or longer
→ channel confirmed jammed

$$Q_p = 0.7 \times PDR + 0.3 \times (1 - J_{ch})$$

01 Path 1: GS1 → GS2 direct link

02 Path 2: GS1 → SAT (2.4 GHz)
SAT → GS2 (433 MHz)

Switch path when the new path's Qp exceeds the current one by 0.15 or more and at least 200 ms has elapsed since the last switch (hysteresis)

FreeRTOS System

Core 0 — Signal Reception & Transmission

2.4 GHz Transceiver

Receives nRF24 packets from GS1 and stores them in a queue for processing.

433 MHz Transceiver

Transmits the received data as packets and re-transmits to GS2 on 433 MHz.

IPC (Inter-Process Communication)

(between two cores)

→ **Receive signal (rfRxQueue)**

From Core 0 to Core 1

← **Transmit command (txQueue)**

From Core 1 to Core 0

Shared State Lock (stateMutex)

Prevent simultaneous modification of the same value by both cores.

Core 1 — Monitoring & Decision Making

Jamming Detection — 50 Hz

Monitor signal strength and packet loss to determine the presence of jamming in real time.

Path Selection — 20 Hz

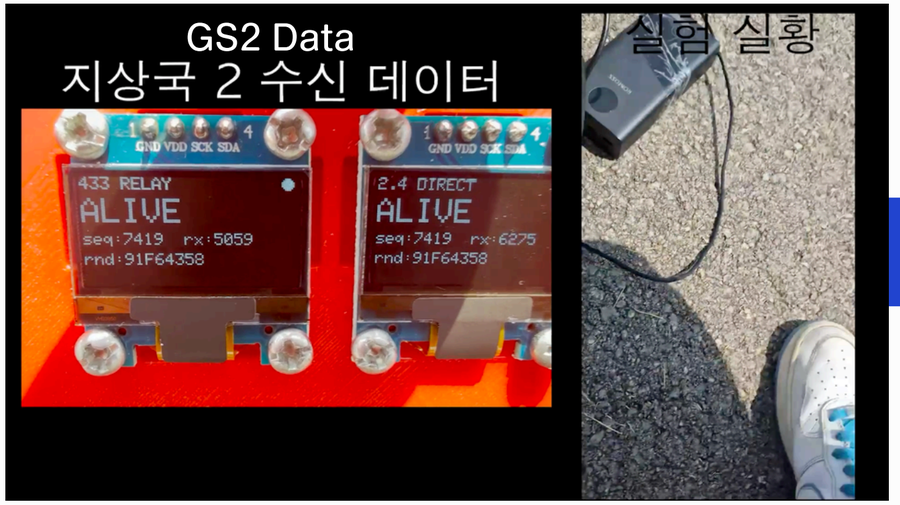
Select the optimal communication path between 2.4 GHz and 433 MHz.

Operation Management — 10 Hz

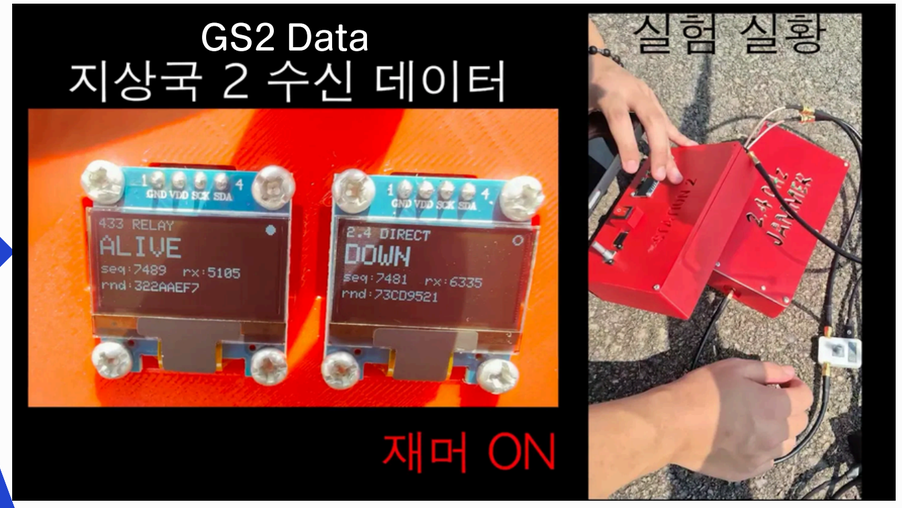
Manage standby → monitoring → jamming detection → relay → recovery state transitions.

TELECOM LOG

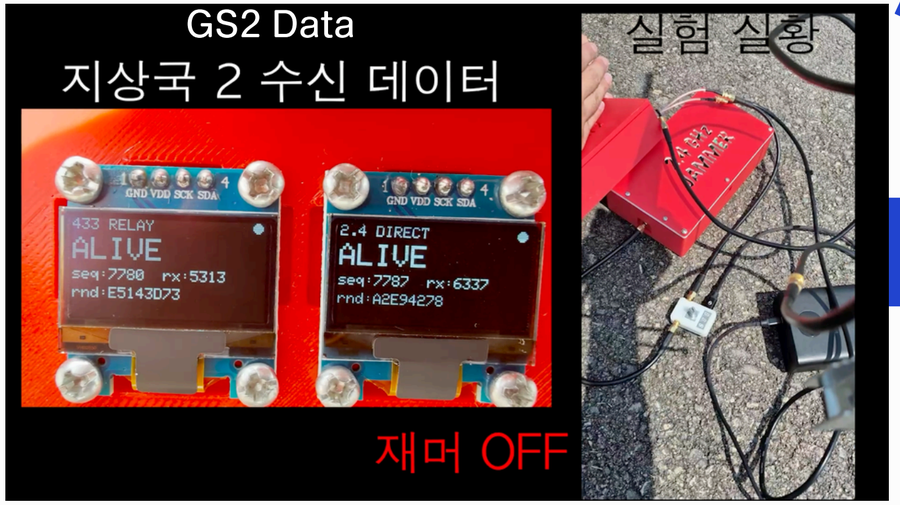
① 433: 7419 / 2.4: 7419, rnd Same



② 433: 7489 / 2.4: 7481 Freeze



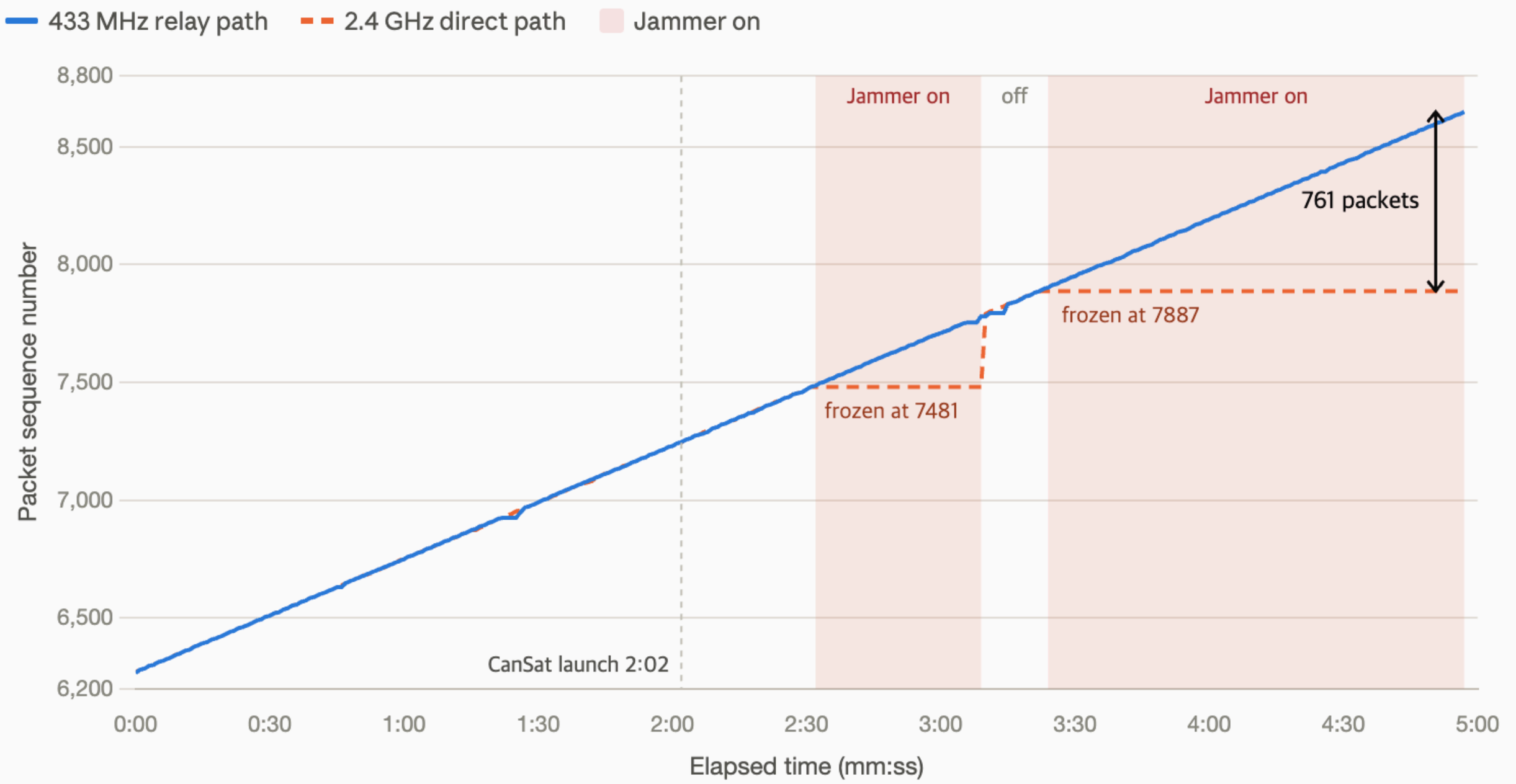
③ 433: 7780 / 2.4: 7787, Normal



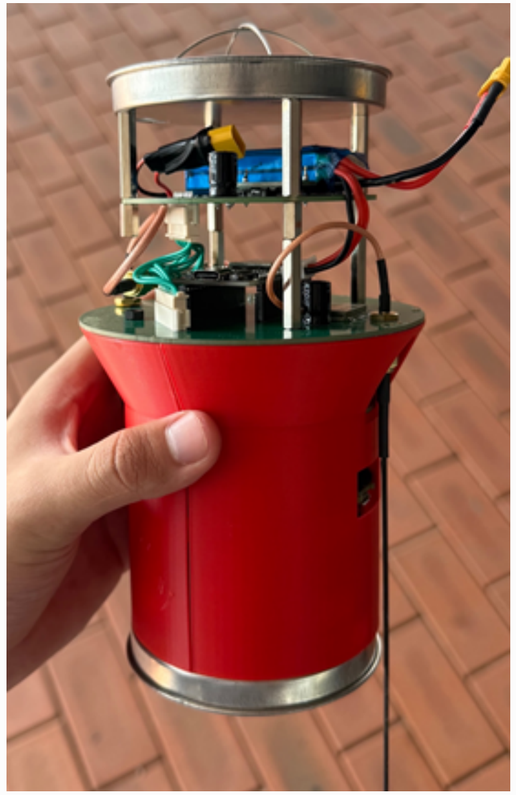
④ 433: 8516 / 2.4: 7887 Freeze



TELECOM LOG


433 MHz relay path
2.4 GHz direct path
Jammer on
Jammer on
off
Jammer on
761 packets
frozen at 7887
frozen at 7481
CanSat launch 2:02
Packet sequence number
8,800
8,500
8,000
7,500
7,000
6,500
6,200
0:00
0:30
1:00
1:30
2:00
2:30
3:00
3:30
4:00
4:30
5:00
Elapsed time (mm:ss)

Research Significance & Future Directions



- Research Significance
  - Validation of anti-jamming technology through satellite relay communication
  - Confirmation that band switching alone can keep the link alive
  - A low-cost architecture applicable to jamming countermeasures for civilian and drone systems
- Future Directions
  - Building a jamming-responsive system through improved satellite-to-ground station protocols
  - Research on spoofing prevention through cross-validation of received signals



